Data-Aided Channel Estimation for OTFS Systems With a Superimposed Pilot and Data Transmission Scheme

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***Abstract*—The recently developed orthogonal time frequency space (OTFS) modulation has shown its capability of coping with the fast time-varying channels in high-mobility environments. In particular, OTFS modulation gives rise to the sparse representation of the delay-Doppler (DD) domain channel model. Hence, one can an enjoy accurate channel estimation by adopting only one pilot symbol. However, conventional OTFS channel estimation schemes require the deployment of guard space to avoid data-pilot interference, which inevitably sacrifices the spectral efficiency. In this letter, we develop a data-aided channel estimation algorithm for a superimposed pilot and data transmission scheme, which can improve the spectral efficiency. To accurately estimate the channel and detect the data symbols, we coarsely estimate the channel based on the pilot symbol, followed by an iterative process which detects the data symbols and refines the channel estimates. Simulation results show that the bit error rate (BER) performance based on the proposed method can approach the baseline scheme with perfect channel estimation.**

***Index Terms*—OTFS modulation, channel estimation, pilot-data superposition.**

I. Introduction

ORTHOGONAL frequency division multiplexing (OFDM) modulation, which has been widely adopted in existing communication systems, offers the advantages of multi-path resilience and low-complexity equalization [1]. Nevertheless, in high-mobility scenarios, the communication channel may experience not only frequency selectivity introduced by the multi-path effect, but also time selectivity due to the Doppler spreads, i.e., doubly selectivity. Since the orthogonality between sub-carriers are affected by doubly selective channels, OFDM suffers from performance degradation, which is challenging for guaranteeing reliable communications [2]. Therefore, new modulation schemes are highly desirable for supporting future mobile systems with high-mobility.

Recently, Hadani *et al.* developed a new modulation scheme, namely orthogonal time frequency spacing (OTFS), which has shown superiority over the classic OFDM scheme in doubly selective channels [3]. The OTFS modulation scheme multiplexes the data symbols in the delay-Doppler (DD) domain instead of the traditional time-frequency (TF) domain. Specifically, the OTFS modulation allows the doubly selective channel to be represented by only a few parameters in the DD domain, providing a relatively stable and sparse channel model, which facilitates the receiver design. Moreover, relying on a two-dimensional (2D) transformation, all information symbols in an OTFS frame are spread over the whole TF domain such that full TF diversity can be exploited.

To unleash the potential of OTFS systems, various investigations have been conducted in the literature. For example, the general framework for OTFS modulation and demodulation was devised in the pioneering paper [4] based on pulse-shaping waveforms. Then, the authors of [5] showed that OTFS can be implemented OTFS using existing OFDM systems by adding some signal processing blocks. The multiple access schemes based on multiple-input multiple-output (MIMO)-OTFS systems have been discussed in [6], [7], which have shown the potential of adopting OTFS modulation for supporting multiple users. To realize OTFS communication, efficient data detection is required. As a result, several works in the literature have discussed the OTFS symbol detection approaches, e.g., minimum mean square error (MMSE) estimation [8], Gaussian message passing [9], approximate message passing [10], and variational Bayes [11], which have different levels of computational complexity and error performance. As a prerequisite for symbol detection, accurate estimation of the channel parameters is of the utmost importance. In [12], OTFS channel estimation based on one pilot symbol was proposed. For highly-accurate estimation, [12] requires the insertion of guard symbols to avoid the interference between the pilot and data symbols. However, the adoption of guard space will degrade the spectral efficiency since only fewer data symbols can be carried in one OTFS frame, especially for the low latency communication scenarios requiring small frame sizes. Therefore, an OTFS channel estimation scheme with enhanced spectral efficiency needs to be investigated.

In this letter, we develop a data-aided channel estimation algorithm for superimposed data and pilot transmission of OTFS systems. Compared with existing approaches placing guard symbols in the DD domain, the proposed method in our paper utilizes whole frame for data transmission and a single pilot symbol is superimposed to the data symbol. As a result, more data symbols can be carried, achieving a higher spectral efficiency compared to existing approaches. Since the pilot and data symbols are superimposed, the interference between

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data symbols and pilot symbol is unavoidable. To eliminate the interference, we propose a low-complexity iterative channel estimation, interference cancellation, and data detection algorithm. The channel parameters are first coarsely estimated using a threshold by taking the interference into account. Then, the estimated channel is used for data detection. In particular, we commence from the maximum *a posteriori* (MAP) symbol detection and design a sum-product algorithm (SPA)-based detector. After performing interference cancellation via the detected data symbols, the channel estimation is refined with an appropriately modified threshold. Through the proposed iterative process, the channel parameters can be accurately estimated. Different from a related work adopting a superimposed pilot-data placement scheme [13], the proposed method in our paper still utilizes only one pilot symbol and simply obtains the channel estimate using the updated threshold, which has a relatively lower implementation complexity than the MMSE channel estimation in [13]. We demonstrate the effectiveness of the proposed algorithm through simulations and show that relying on the SPA-based detector, the proposed algorithm can achieve a similar bit error rate (BER) performance of the classic approach while improves the spectral efficiency.

Notations: The superscripts $(\cdot)^{\mathrm{T}}$, $(\cdot)^*$, and $(\cdot)^{\mathrm{H}}$ denote the transpose, conjugate, and the Hermittian operations, respectively; $\propto$ represents both sides of the equation are multiplicatively connected to a constant; $|\cdot|$ denotes the modulus of a complex number or the cardinality of a set; $(\cdot)_N$ denotes the modulo operation with divisor N; $\mathcal{CN}(m, v)$ denotes a complex Gaussian distribution with mean m and variance v.

II. System Model

In the following, we discuss the basic OTFS transceiver structure based on [4]. A sequence of information bits is mapped to MN data symbols belonging to a constellation set $\mathcal{A}$. The data symbols are placed in MN two-dimensional (2D) DD domain grids, i.e., $x[k, l]$, where the sub-indices $0 \le k \le N-1$ and $0 \le l \le M-1$ denote the Doppler and delay indices, respectively. The DD domain symbols are mapped to the TF domain via inverse symplectic finite Fourier transform (ISFFT),

$$X[n,m] = \frac{1}{\sqrt{MN}} \sum_{k=1}^{N-1} \sum_{l=0}^{M-1} x[k,l] e^{j2\pi\left(\frac{nk}{N} - \frac{ml}{M}\right)}, \quad (1)$$

where $0 \le n \le N-1$ and $0 \le m \le M-1$ are the indices for time and subcarrier, respectively. Next, a multi-carrier modulator is used for transforming the TF domain symbols to a time domain continuous signal as

$$s(t) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} X[n,m] e^{j2\pi m\Delta_f (t-nT)} g_{\mathrm{tx}}(t-nT), \quad (2)$$

where $g_{\mathrm{tx}}(t)$ is the pulse shaping filter, Δ_f and T are the subcarrier spacing and symbol duration, respectively.

After transmitting through a multi-path time-varying channel, the received signal is given by

$$r(t) = \int\int h(\tau,\nu) e^{j2\pi\nu(t-\tau)} s(t-\tau) d\tau d\nu + \omega(t), \quad (3)$$

where $\omega(t)$ is the additive white Gaussian noise process with power spectral density (PSD) N_0. The channel representation in the DD domain is given by [4]

$$h(\tau,\nu) = \sum_{i=1}^{P} h_i \delta(\tau-\tau_i)\delta(\nu-\nu_i), \quad (4)$$

with P the number of resolvable reflectors, h_i the channel gain, τ_i the delay, and ν_i the Doppler of the i-th path, respectively.

At the receiver side, the TF domain received symbol $Y[n, m]$ is obtained by performing the conventional multi-carrier demodulation. Finally, the TF domain symbol $Y[n, m]$ is transformed to the DD domain via the symplectic finite Fourier transform (SFFT), i.e.,

$$y[k,l] = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y[n,m] e^{-j2\pi\left(\frac{kn}{N} - \frac{lm}{M}\right)}. \quad (5)$$

For notational purpose, we define $\tau_i = l_{\tau_i} \frac{1}{M\Delta_f}$ and $\nu_i = k_{\nu_i} \frac{1}{NT}$, where l_{τ_i} and k_{ν_i} are the delay index and Doppler index associated with the i-th path, respectively.[1] Having the DD domain received signal $y[k, l]$ and transmitted symbol $x[k, l]$, we are able to represent the DD domain input-output relationship as

$$y[k,l] = \sum_{k'=0}^{N-1} \sum_{l'=0}^{M-1} x[k',l'] h_w\left[(k-k')_N, (l-l')_M\right] + \omega[k,l], \quad (6)$$

where $h_w[k, l]$ denotes the DD domain effective channel and $\omega[k, l]$ is the DD domain noise sample. Since the transformation from the time domain to the DD domain is orthogonal, $\omega[k, l]$ is a zero-mean Gaussian noise term with PSD N_0.

III. Data-Aided Channel Estimation for OTFS Modulation

A. Pilot Placement

For the classic approach [12], given the maximum delay index and Doppler index given by $l_{\max}$ and $k_{\max}$, channel estimation can be done through the following pilot placement in the DD domain

$$x[k,l] = \begin{cases} \text{pilot } x_p, & k = k_p,\ l = l_p, \\ 0, & k \in [k_p - 2k_{\max}, k_p + 2k_{\max}] \backslash k_p, \\ & l \in [l_p - l_{\max}, l_p + l_{\max}] \backslash l_p. \\ \text{datasymbol}, & \text{otherwise}. \end{cases}$$

We can see that a total number of $N_{\mathrm{guard}}+1$ DD domain grids are used for OTFS channel estimation. Hence, the spectral efficiency can be defined as $\eta = \frac{\log|\mathcal{A}|MN - N_{\mathrm{guard}} - 1)}{M\Delta_f \cdot NT}$ bits/s/Hz, where $M\Delta_f$ and NT denote the frequency and time resources occupied by the OTFS frame. Note that we have $\Delta_f T = 1$, which simplifies the expression of η as $\eta = \frac{\log|\mathcal{A}|(MN - N_{\mathrm{guard}} - 1)}{MN}$ bits/s/Hz. For a high-mobility scenario with a high Doppler shift, we need a large value of N_{guard} to avoid the interference between data symbols and pilot symbols. Meanwhile, in the future wireless communications, latency requirement is highly stringent. In other words, a relatively short OTFS frame duration is expected, which places a strong constraint on the size of the OTFS frame. In such cases, the spectral efficiency is severely affected. Hence, in

[1] In this letter, only integer Doppler case is considered. In the presence of fractional Doppler, the window design scheme proposed in [14] can be exploited to maintain the sparsity of the DD domain effective channel.

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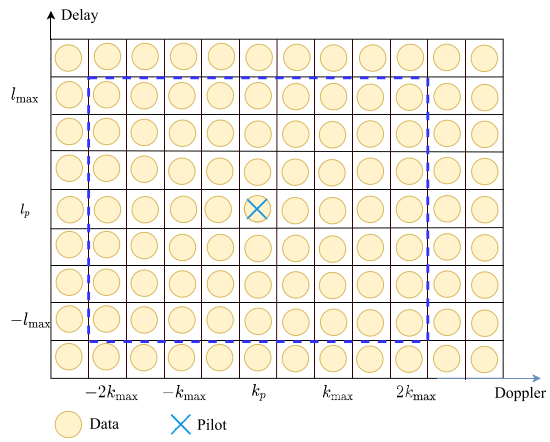

Fig. 1. Pilot placement for data-aided channel estimation scheme.

this section, we propose a data-aided scheme that allows the whole OTFS frame for data transmission. In particular, the DD domain symbol arrangement is depicted in Fig. 1, where the area bounded by the blue dashed line denotes the guard space adopted in the classic OTFS channel estimation.

As illustrated in Fig. 1, we propose the following superimposed pilot and data transmission scheme

$$\tilde{x}[k,l] = \begin{cases} x_p + x[k_p, l_p], & k = k_p,\ l = l_p, \\ x[k,l], & \text{otherwise}. \end{cases} \tag{7}$$

We denote the pilot symbol energy by $E_p = |x_p|^2$ and the average data symbol energy by $E_s = \mathbb{E}\{|x[k,l]|^2\}$. We can observe that no dedicated DD domain grids are occupied for OTFS channel estimation. However, since the pilot symbol x_p is superimposed with the data symbol $x[k_p, l_p]$, the interference is unavoidable. Due to the intentional interference induced by data symbols, the received samples related to the pilot can be rewritten as

$$y[k,l] = x_p h_w[(k-k_p)_N, (l-l_p)_M] + \mathcal{I}_{k,l} + w[k,l], \tag{8}$$

where $\mathcal{I}_{k,l}$ denotes the interference introduced by the data symbols, given by

$$\mathcal{I}_{k,l} = \sum_{k'=k-k_{\max}}^{k-k_{\max}} \sum_{l'=l-l_{\max}}^{l} x[k',l'] h_w[(k-k')_N, (l-l')_M].$$

Due to the existence of data-pilot interference, the channel parameters can not be simply obtained using the threshold $\gamma = 3\sqrt{N_0}$ for channel estimation. In the following section, we will propose an iterative data-aided channel estimation algorithm for OTFS communication systems.

B. The Proposed Channel Estimation Scheme

The block diagram of the proposed joint OTFS channel estimation and data detection scheme is given in Fig. 2. The channel is coarsely estimated using the received signal $y[k, l]$, followed by a sum-product algorithm (SPA)-based symbol detection. Then we can cancel out the interference from the received signal $y[k, l]$ based on the detected symbols and refine the estimation of channel parameters. The coarse channel estimation is done by comparing the received signal with a modified threshold γ, which takes account of the effect of noise as well as the interference $\mathcal{I}_{k,l}$. Based on the expression

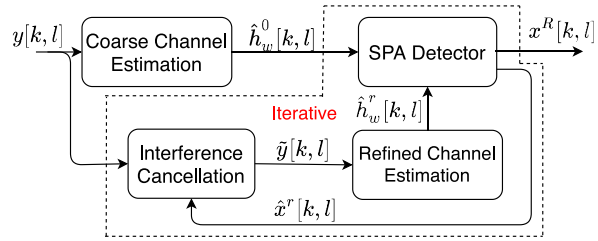

Fig. 2. Block diagram for the proposed data-aided OTFS channel estimation.

of h_w [4], we rewrite the interference term $\mathcal{I}_{k,l}$ as

$$\mathcal{I}_{k,l} = \sum_{i \in \mathcal{Q}_{k,l}} h_i x[(k-k_{\nu_i})_N, (l-l_{\tau_i})_M] e^{-j2\pi \frac{k_{\nu_i} l_{\tau_i}}{NM}}, \tag{9}$$

where $\mathcal{Q}_{k,l}$ denotes the set containing indices of all data symbols contributed to sample $y[k, l]$. In particular, the pilot is interfered by P data symbols, therefore $|\mathcal{Q}_{k,l}| = P$. Having (9) in hand, we can calculate the interference energy, expressed as

$$\begin{aligned} \mathbb{E}\left\{|\mathcal{I}_{k,l}|^2\right\} &= \sum_{i \in \mathcal{Q}_{k,l}} \mathbb{E}\left\{|h_i|^2\right\} \mathbb{E}\left\{|x[(k-k_{\nu_i})_N, (l-l_{\tau_i})_M]|^2\right\} \\ &+ \sum_{i,i' \in \mathcal{Q}_{k,l}} \mathbb{E}\{h_i h_{i'}^*\} \mathbb{E}\{x[(k-k_{\nu_i})_N, (l-l_{\tau_i})_M] \\ &\cdot x^*\left[(k-k_{\nu_{i'}})_N, (l-l_{\tau_{i'}})_M\right]\} e^{-j2\pi \frac{k_{\nu_i} l_{\tau_i} - k_{\nu_{i'}} l_{\tau_{i'}}}{NM}}, \end{aligned} \tag{10}$$

provided that data symbols are independent from the channel coefficients. Assuming that the channel coefficients are independent for different paths $\mathbb{E}\{h_i h_{i'}^*\} = 0$, we can further simplify (10) as

$$\mathbb{E}\left\{|\mathcal{I}_{k,l}|^2\right\} = \sum_{i \in \mathcal{Q}_{k,l}} \mathbb{E}\left\{|h_i|^2\right\} E_s. \tag{11}$$

In this letter, we assume that the channel gains obey the exponential power delay profile. Note that the channel power gains are in general normalized, yielding $\sum_{i=1}^{P} \mathbb{E}\{|h_i|^2\} = 1$. Therefore, the energy of the interference term is given by $\mathbb{E}\{|\mathcal{I}_{k,l}|^2\} = E_s$. Consequently, we have the modified threshold for coarse channel estimation, given by

$$\gamma = 3\left(\sqrt{N_0 + E_s}\right). \tag{12}$$

If $|y[k,l]| \geq \gamma$, the estimate of the effective DD domain channel can be obtained as

$$\hat{h}_w\left[(k-k_p)_N, (l-l_p)_M\right] = y[k,l]/x_p. \tag{13}$$

By exploiting the 2D convolution property of the OTFS system, one can efficiently determine the estimate of channel gains h_i as well as the delay and Doppler indices associated with the DD channel.

After obtaining the channel estimates, we can detect the OTFS data symbols using the estimated channel $\hat{h}_w$. By stacking all received samples $y[k, l]$ as $\mathbf{y}$, the symbol-wise optimal MAP detector is given by

$$\hat{x}[k,l] = \arg\min_{x[k,l] \in \mathcal{A}} p(x[k,l]|\mathbf{y}), \tag{14}$$

where $p(x[k, l]|\mathbf{y})$ denotes the marginal *a posteriori* probability of $x[k, l]$ given the received signals $\mathbf{y}$. Note that calculating $p(x[k, l]|\mathbf{y})$ through direct marginalization is computational

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inefficient. Here, we resort to the sum-product algorithm (SPA) [15] for deriving the marginal probability $p(x[k, l]|\mathbf{y})$. According to the Bayes' theorem, we write

$$p(x[k,l]|\mathbf{y}) \propto p(\mathbf{y}|x[k,l])p(x[k,l]). \tag{15}$$

For the considered channel model, a data symbol will contribute to total P received samples. Without loss of generality, we define $\mathbb{Y}_{k,l}$ as the set of all received samples that are related to $x[k, l]$, where $\mathbb{Y}_{k,l}^{[i]} = y[(k+k_{\nu_i})_N, (l+l_{\tau_i})_M]$ denotes the i-th element of $\mathbb{Y}_{k,l}$. Consequently, we can rewrite $p(\mathbf{y}|x[k, l])$ as

$$\begin{aligned} p(\mathbf{y}|x[k,l]) &\propto \prod_{i=1}^{P} p_{\mathbb{Y}_{k,l}^{[i]} \to x[k,l]} \\ &= \prod_{i=1}^{P} \sum_{x[k',l'] \in \mathbb{X}_{k,l}^{[i]}} p\Big(\mathbb{Y}_{k,l}^{[i]} | x[k,l], \mathbb{X}_{k,l}^{[i]}\Big) p\Big(x\Big[k',l'\Big] \to \mathbb{Y}_{k,l}^{[i]}\Big), \end{aligned} \tag{16}$$

where $\mathbb{X}_{k,l}^{[i]}$ denotes all data symbols related to $\mathbb{Y}_{k,l}^{[i]}$ except for $x[k, l]$, $p_{\mathbb{Y}_{k,l}^{[i]} \to x[k,l]}$ is an auxiliary probability defining the probability of $x[k, l]$ conditioned on $\mathbb{Y}_{k,l}^{[i]}$. We keep in mind that the auxiliary probabilities are equivalent to the messages on factor graph which satisfy $p_{\mathbb{Y}_{k,l}^{[i]} \to x[k,l]} p_{x[k,l] \to \mathbb{Y}_{k,l}^{[i]}} = p(x[k,l]|\mathbf{y})$. Then, $p_{x[k,l] \to \mathbb{Y}_{k,l}^{[i]}}$ can be obtained by substituting (16) into (15) and dividing by $p_{\mathbb{Y}_{k,l}^{[i]} \to x[k,l]}$. In particular, we have

$$p_{x[k,l] \to \mathbb{Y}_{k,l}^{[i]}} = p(x[k,l]) \prod_{i' \neq i} p_{\mathbb{Y}_{k,l}^{[i']} \to x[k,l]}. \tag{17}$$

It can be observed that the above detection is done by performing summation and product operations, referred to as the SPA-based detection. After OTFS symbol detection, we have the symbols given by $\hat{x}[k, l]$. Then we are able to cancel out the interference induced by the data symbols based on the estimated channel and the obtained symbols, given by

$$\tilde{y}[k,l] = y[k,l] - \sum_{k'} \sum_{l'} \hat{x}[k',l'] \hat{h}_w \Big[\Big(k-k'\Big)_N, \Big(l-l'\Big)_M\Big]. \tag{18}$$

If the interference is perfectly cancelled, then the residual term $\tilde{y}[k, l]$ contains only pilot information and the noise. In this case, we can use the threshold in the classic channel estimation method for determining the DD domain channel. Nevertheless, due to the impact of noise, the cancellation is generally imperfect. Therefore, we employ a scheme that iteratively performs data detection and channel estimation. It is expected that better channel estimation results will enhance the data detection performance and in turn we have a more accurate interference cancellation for performing the threshold-based channel estimation.

In the r-th iteration of the proposed iterative algorithm, the data symbols $\hat{x}^r[k, l]$ are detected using the estimated channel in the previous iteration, i.e., $\hat{h}_w^{r-1}[k, l]$. Then both $\hat{x}^r[k, l]$ and $\hat{h}_w^{r-1}[k, l]$ are adopted for interference cancellation followed by a threshold-based refined channel estimation. In particular, the threshold γ needs to be updated during the process. In general, the number of channel paths is also unknown to the receiver. As a compromise approach, we consider a relatively large number P' to avoid the misdetection of the channel paths with small gains. Assuming that we have identified J paths out of total P' paths in the $(r-1)$-th iteration, after interference cancellation, there are still interference corresponding to the rest $P' - J$ paths. The energy for the residual interference can be roughly obtained as $\frac{P'-J}{P'} E_s$. Consequently, the new threshold for refined channel estimation at the r-th iteration is given by

$$\gamma^r = 3\left(\sqrt{N_0 + \frac{(P'-J)E_s}{P'}}\right). \tag{19}$$

Then we can derive the refined channel estimates at the r-th iteration for continuing the iterative process. After running R iterations, the channel estimates as well as the detected data symbols are obtained. In general, the iterative process can be terminated if the performance gain by running more iterations is marginal.

As the whole OTFS frame is used for transmitting the data symbols, it can be calculated that the spectral efficiency of the proposed algorithm is $\eta = \frac{\log|\mathcal{A}|(MN)}{MN} = \log|\mathcal{A}|$ bits/s/Hz. Compared to the classic channel estimation, the spectral efficiency has been improved by $\frac{MN}{MN - N_{\text{guard}} - 1} \times 100\%$, which is significant when the size of the transmitted OTFS frame is relatively small.

IV. Simulation Results

We consider an OTFS frame with $N = 30$ and $M = 128$ indicating there are 30 time slots and 128 subcarriers in the TF domain. The operating carrier frequency is $f_c = 3$ GHz and the subcarrier spacing is 7.5 kHz. The maximum moving speed in the scenario is $v = 150$ km/h, yielding a maximum Doppler index with $k_{\max} = 5$, and the maximum delay index is $l_{\max} = 10$. The number of channel paths is $P = 4$ and the associated delay and Doppler indices for each channel path are randomly chosen from $[0, l_{\max}]$ and $[-k_{\max}, k_{\max}]$, respectively. The channel gain is generated according the distribution $\mathcal{CN}(0, \frac{1}{P})$, which is commonly used in the literature [9]. The signal-to-noise ratio (SNR) for the transmission is defined as $\text{SNR} = \frac{E_s}{N_0}$. We assume that the pilot signal-to-noise ratio is $\text{SNR}_p = \frac{E_p}{N_0}$. Finally, quadrature phase shift keying (QPSK) modulation is used for the bit mapping and the maximum number of iterations for data detection and channel estimation is $R = 3$.

In Fig. 3, we illustrate the normalized mean square error (NMSE) performance, i.e., $\text{NMSE}_h = \frac{\|\hat{h}_w - h_w\|^2}{\|h_w\|^2}$, of the proposed data-aided channel estimation algorithm and the classic approach [12] versus the SNR. We assume that $\text{SNR}_p = 40$ dB. Due to the interference between data symbols and the pilot symbol, the coarse channel estimation by regarding all interference as noise suffers from significant performance loss. Since we use a fixed pilot energy, with the increase of the SNR, the interference to the pilot symbol becomes more severe, leading to an increase of the NMSE of the proposed algorithm. Nevertheless, by performing the proposed iterative algorithm, the NMSE performance can be significantly improved after only 3 iterations, which can approach the classic method [12] with guard space (Classic W). However, our algorithm can achieve a spectral efficiency of $\eta = 2$ bits/s/Hz while the classic approach has $\eta = 1.77$ bits/s/Hz. We can see that the superimposed transmission scheme achieves an improvement of 13.0% of the spectral

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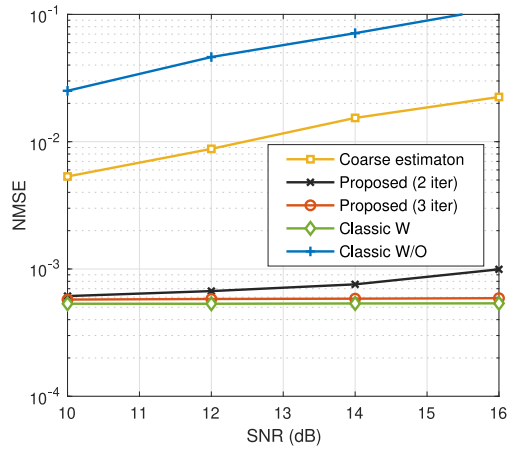

Fig. 3. NMSE performance versus SNR.

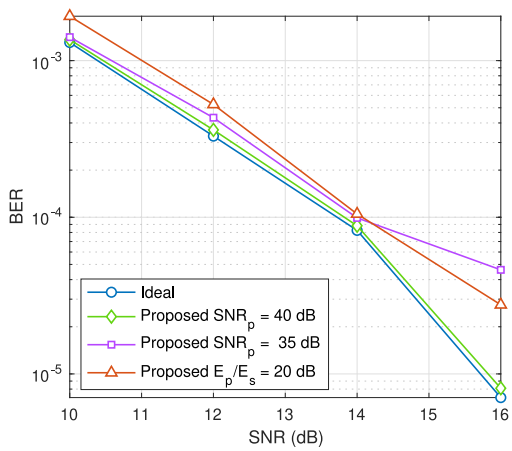

Fig. 4. BER performance versus SNR.

efficiency. This improvement becomes more attractive when supporting multiple users where more guard space is required for avoiding inter-user interference. In this figure we also plot the channel estimation result based on threshold $3\sqrt{N_0}$ without adopting guard space (Classic W/O), which has the same spectral efficiency of the proposed algorithm. However, it can be seen that neglecting the interference severely degrades the channel estimation performance, as shown in Fig. 3.

In Fig. 4, we illustrate the bit error rate (BER) performance based on the SPA detector. Two values of pilot SNR, i.e., 35 dB and 40 dB are considered. We can see that for the case of $\mathrm{SNR}_p = 40$ dB, the BER performance based on the proposed algorithm almost coincides with that of the ideal case, where the channel parameters are perfectly known to the detector. With $\mathrm{SNR}_p = 35$ dB, the detection performance degrades in the high SNR regime, which is due to the channel estimation error caused by a relatively high interference energy to the fixed transmission pilot symbol energy. A viable approach to mitigate this effect is to adopt an adaptive transmission pilot symbol energy. In Fig. 4, we also illustrate the BER performance of our algorithm by using a pilot-to-data energy ratio of 20 dB. In this case, SNR_p also becomes higher with the increase of SNR, leading to better channel estimation results as well as BER performance. Moreover, compared with the coarse channel estimation, a 2 dB SNR gain can be observed by reducing the NMSE of channel estimation. As for the peak-to-average-power ratio (PAPR), with the symbol power of 15 dB and the pilot power of 35 dB, the PAPR for the proposed symbol placement scheme is 21.1 dB compared to 21.07 dB for the conventional scheme with guard space. Nevertheless, the spectral efficiency can be increased by 13.00%, at the cost of only 0.14% PAPR increase in dB.

V. Conclusion

This letter developed a data-aided channel estimation algorithm for OTFS modulation. Compared with the classic OTFS channel estimation scheme, we no longer adopt the guard space and consider a superposition of pilot and data symbols, which can significantly improve the spectral efficiency, especially for the low-latency communication scenario where the frame size can not be too large. To mitigate the interference between the data symbols and pilot, we proposed an iterative data detection, interference cancellation, and channel estimation algorithm. The data symbols are detected based on the estimated channel, which is used for cancelling the pilot-data interference and refining the channel estimates. Simulation results showed that the performance relying on the proposed scheme can approach that of the classic algorithm, while improving the spectral efficiency.

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